

Lesson 29: Conditional Probability & Trees — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Two fair coins: $P(\text{two heads})$?
2. Two dice: $P(\text{two sixes})$?
3. Along branches of a tree, do you $\times$ or $+$?
4. Across separate paths, do you $\times$ or $+$?
5. A coin then a die: $P(\text{tails then a 6})$?
6. Does drawing without replacement change later probabilities?

Section 2 — Core practice

7. Bag 4 red, 1 blue (no replacement). $P(\text{red then blue})$?
8. Bag 2 red, 3 blue (no replacement). $P(\text{both blue})$?
9. $P(A \cap B) = 0.2$, $P(B) = 0.5$. Find $P(A|B)$.
10. Independent A, B: $P(A)=0.4$, $P(B)=0.5$. Find $P(A \cap B)$.
11. $P(A \cap B) = 0.18$, $P(B) = 0.3$. Find $P(A|B)$.

12. A coin then a die: $P(\text{heads then even})$?

Section 3 — Challenge & exam-style

13. Box: 5 good, 3 faulty bulbs. Two taken without replacement. Find $P(\text{both good})$.

14. Using the box above, find $P(\text{exactly one faulty})$.

15. Bag 3 red, 2 green (no replacement). Find $P(\text{one of each, in any order})$.

16. $P(A|B) = 0.4$ and $P(B) = 0.5$. Find $P(A \cap B)$.